Recap

Supervised x, y → Rule

Regression

Linear Regression

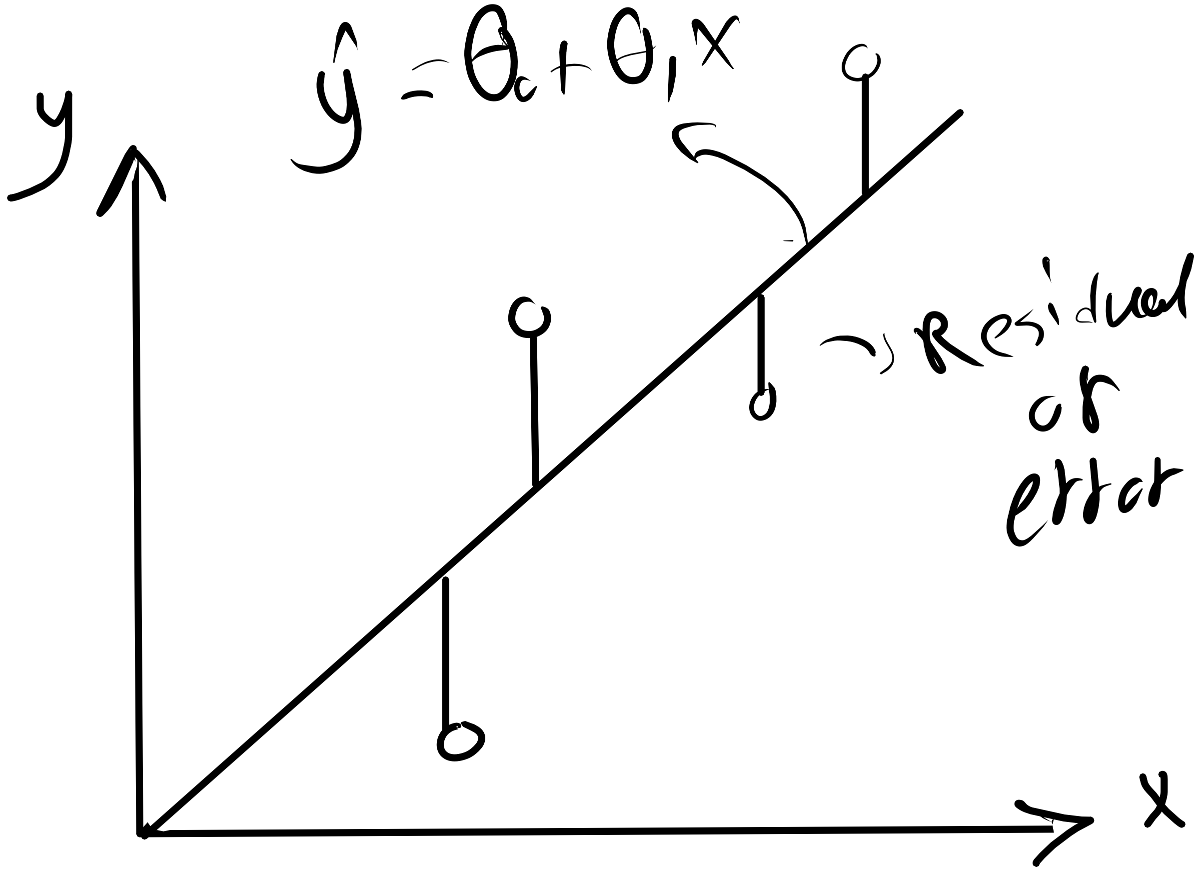
Simple Multiple Polynomial Regularized

Find "Best fit line"

x, y

$$RSS = \sum_{i=1}^{n} (\hat{y} - y_i)^2$$

$$RSS = \sum_{i=1}^{n} \left[(\theta_0 + \theta_1 x_i) - y_i \right]^2$$



$$\theta_0, \theta_1 \longrightarrow RSS \downarrow\downarrow$$

Solve for θ_0, θ_1

Analytical Solution

Ordinary Least Squares (OLS)

$$\theta = (X^T X)^{-1} X^T \cdot y$$

Optimization Solution

Gradient Descent

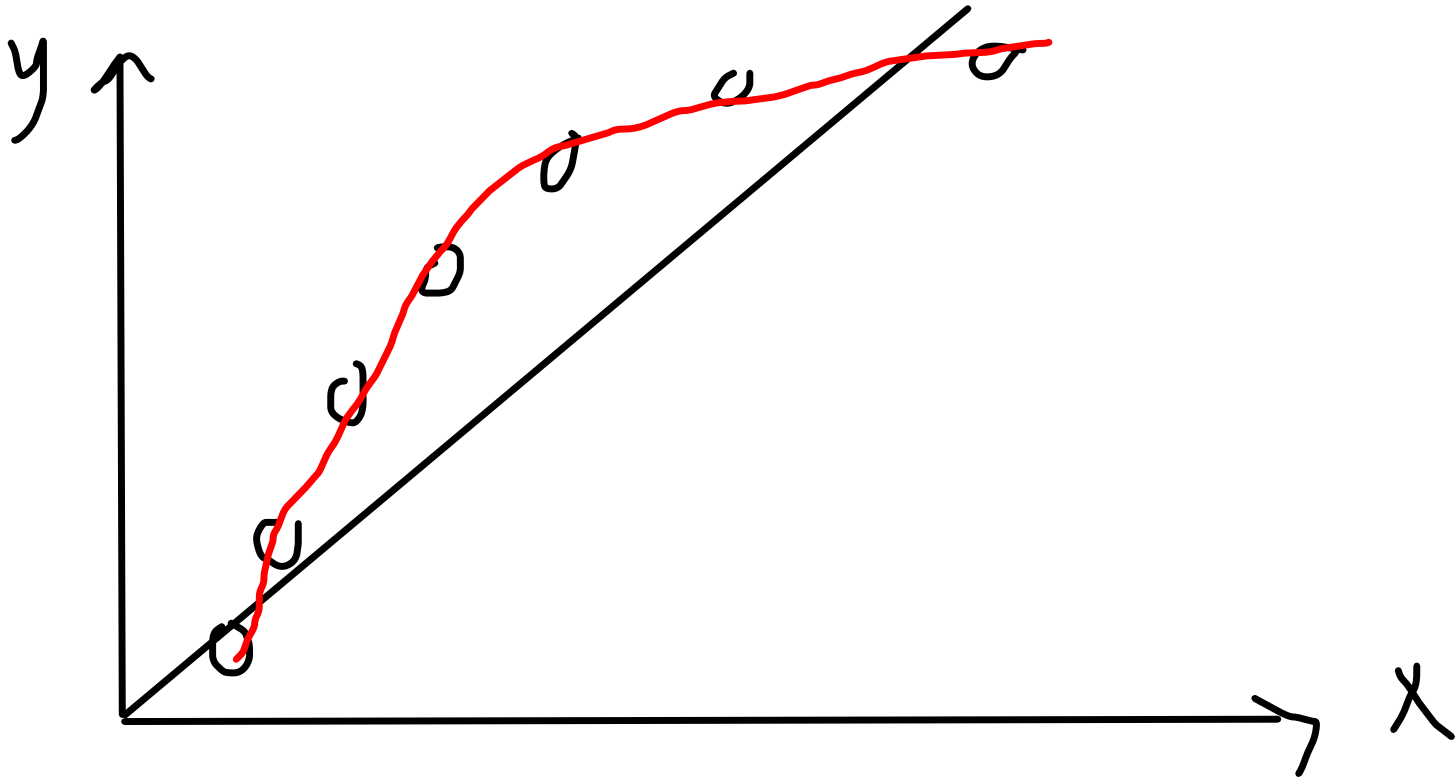
1- Initiate random values for theta_0 and theta_1
2- Calculate Loss
3- Update theta_0 and theta_1

Evaluation Metrics

R2 $\rightarrow (-\infty \rightarrow 1)$

RMSE $= \frac{1}{2n} \sum_{i=1}^{n} \sqrt{(\hat{y} - y_i)^2}$

Polynomial



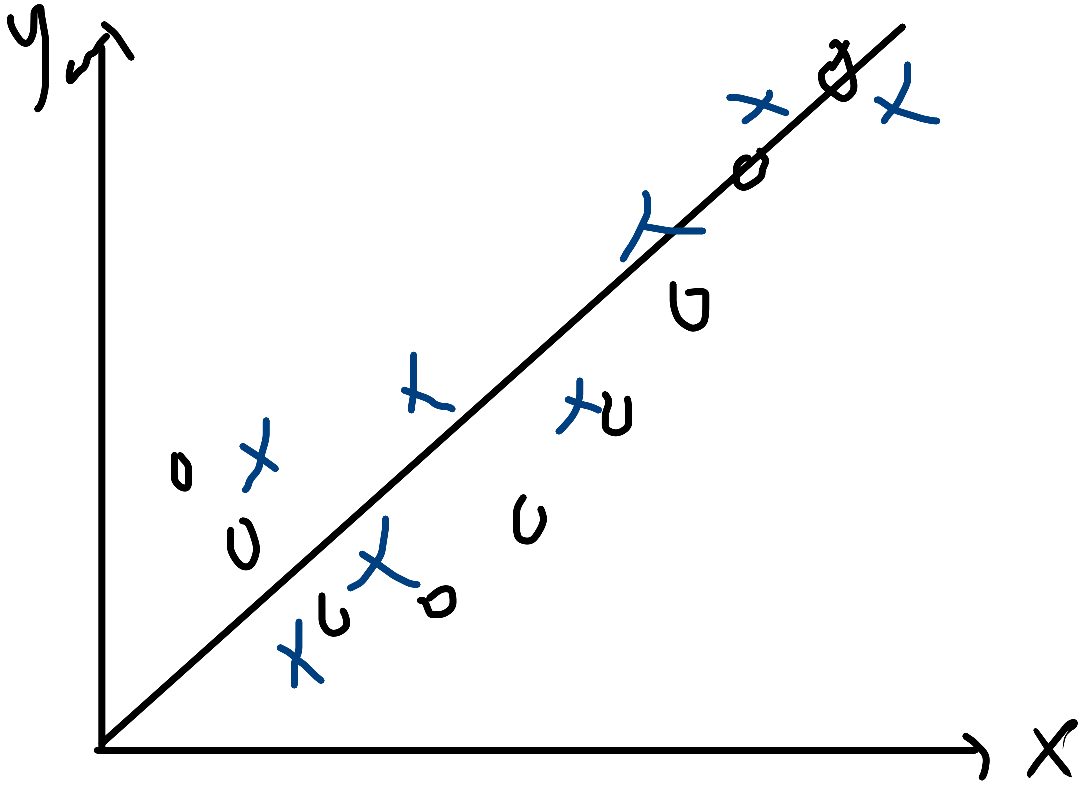
$$y = \theta_0 + \theta_1 x$$

$$y_{poly} = \theta_0 + \theta_1 x + \theta_2 x^2$$

$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

$$y_{poly} = \theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_2 + \theta_4 x_2^2 + \theta_5 x_1 x_2$$

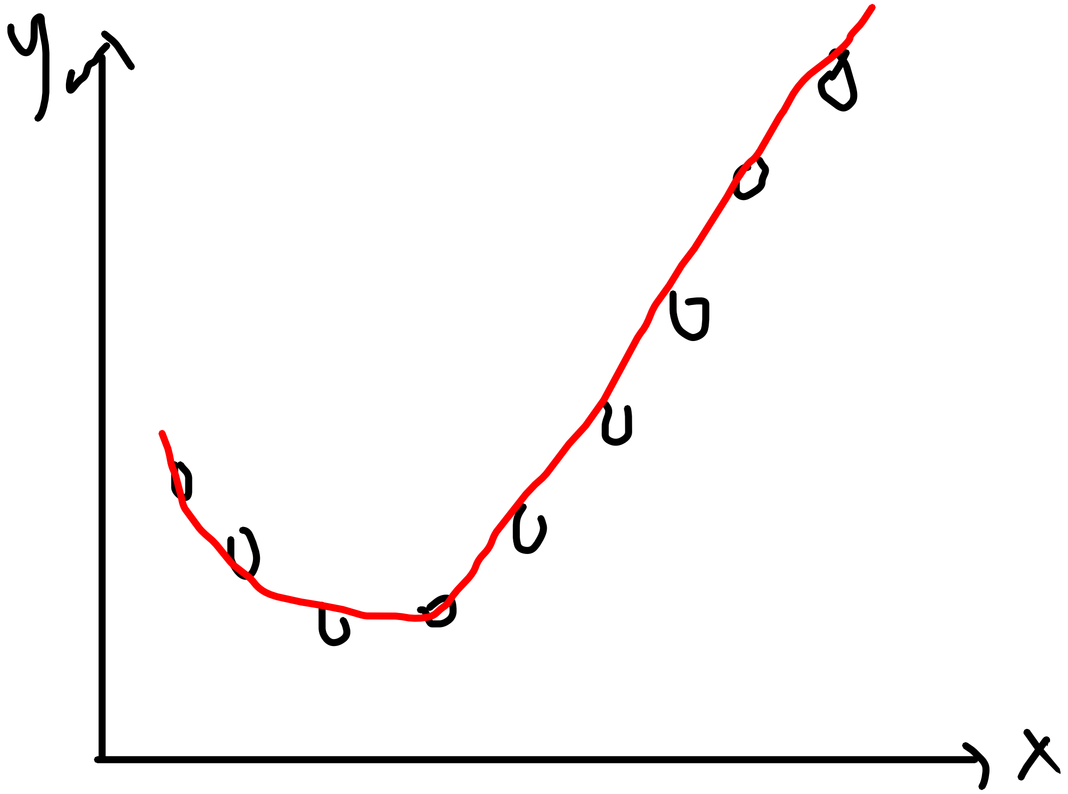
Underfitting VS Overfitting



$y = \theta_0 + \theta_1 x$

Underfitting

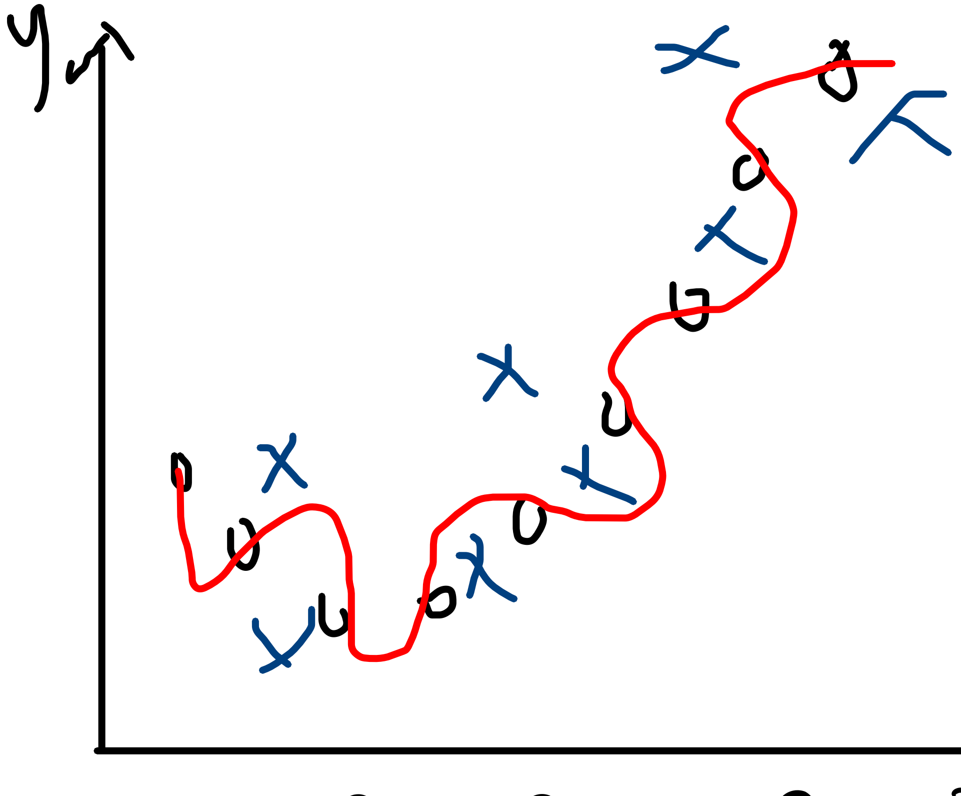
Train R2 = 60%
Test R2 = 50%



$y = \theta_0 + \theta_1 x + \theta_2 x^2$

Best Fit

Train R2 = 90
Test R2 = 85%



$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$

Overfitting

Train R2 = 99%
Test R2 = 60%

Regularization

L1 Regularization (Lasso)

$$J(\theta) = \frac{1}{2n} \sum_{i=1}^{n} \left[(\theta_0 + \theta_1 x_i) - y_i \right]^2 + \lambda \sum_{i=1}^{n} |\theta_i|$$

Regularization

L2 Regularization (Ridge)

$$J(\theta) = \frac{1}{2n} \sum_{i=1}^{n} \left[(\theta_0 + \theta_1 x_i) - y_i \right] + \lambda \sum_{i=1}^{n} (\theta_i)^2$$